

Ethan Rosenfeld

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Education

Duke University, Durham, NC

Expected May 2029

GPA: 4.0

Majors: Mechanical Engineering, Physics

Certificates: Aerospace Engineering

Horace Mann High School, Bronx, NY

June 2025

Henry J. Binder, M.D. '53 Scientific Achievement Award in Physics (12)

Robotics, Computer Science, and Engineering Student of Distinction Award (12)

Work Experience

Mid Island Air Service, *Aircraft Maintenance (Intern)*, Shirley, NY

Summer 2023

- Performed annual and 100-hour inspections alongside airframe and powerplant mechanics on Beechcraft, Cessna, and Piper aircraft.
- Assisted in engine maintenance, including cylinder compression tests, servicing spark plugs, changing oil filters and fluid lines, and replacing and timing magnetos
- Serviced flight control system, seats, and interior carpeting

Extracurricular Activities

DukeAERO, *Propulsion Engineer*, Durham, NC

2025-Present

- As Fluid Systems Team Lead (2026-Present), designed propellant tanks, lines, and valves for DukeAERO's first liquid rocket engine, targeting a 15,000 ft apogee
- Modeled two-phase NOX injector flow and built a full-vehicle performance simulator to size tanks and injectors
- Designed, FEA-validated, and CNC-machined flight hardware, including a custom dynamic piston and tank closures
- Redesigned the team's solid rocket motor nozzle, increasing thrust and specific impulse through first-principles derivation and Ansys Fluent CFD, flying successfully to a 29,400 ft apogee

MIT BWSI Unmanned Aerial System Synthetic Aperture Radar, Cambridge, MA

Summer 2024

- Programmed a Raspberry Pi to collect data from a PulsON 440 radar via UDP
- Designed and interpreted Range-Time-Intensity plots from collected data
- Developed a time-domain backprojection algorithm in Python to create synthetic aperture radar (SAR) maps
- Detected buried landmines using SAR in collaboration with Lincoln Laboratory, the only team of five to do so

FRC Teams 5806, 10452, *Electronics Lead*, Bronx, NY

2023-2025

- Implemented Java-based QR code detection to automatically align the robot for scoring
- Automated multi-step scoring sequences in Java to simplify robot controls
- Designed and organized electrical component layout

Technical Projects

Math and Physics Demonstrations (JavaScript, HTML, CSS)

2024-Present

- Web apps to demonstrate concepts in math and physics
- Projects include vector fields, circuits, electric fields, and matrix transformations.

Electromagnetism (Julia)

2024-2026

- Julia script to calculate and display electric and magnetic fields given a configuration of moving charges

Flight Simulator Asset Design (Photoshop)

2023-2026

- Created various airport sceneries and designed liveries for aircraft.

Technical Skills

Certificates: Private Pilot License, Instrument Rating

Languages: Python, Java, JavaScript, Julia, HTML, CSS

Software: MATLAB, Ansys Fluent, Onshape, Fusion, Solidworks, FEA, Git, Photoshop

Hardware: Raspberry Pi, Arduino